













Validating Causality







S



-

R  +

R.A. -



S



-



C



-



R  +

R.B. -



C



—





PO

PO









PO

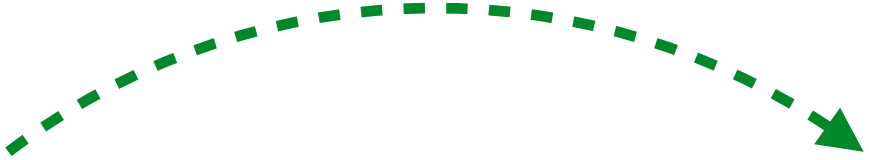
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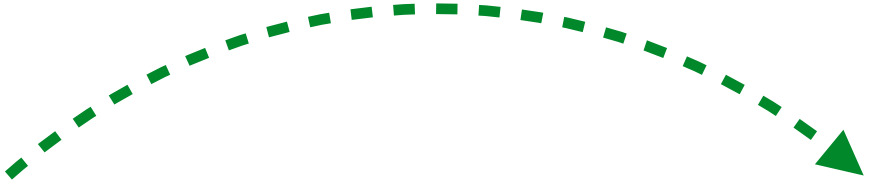
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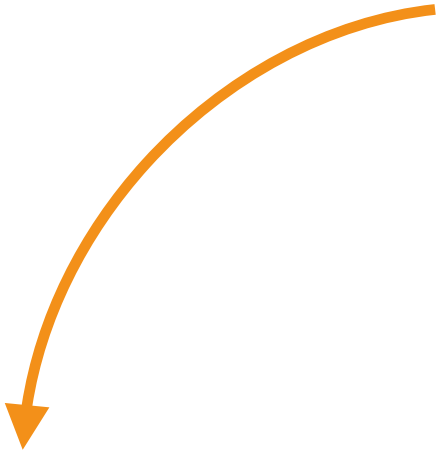
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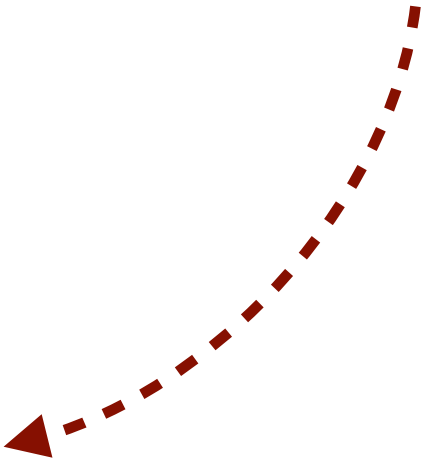








collog



h

b

```
update(src, +100)
```



```
update(src, -100)
```



```
def update(acc: cown[Account], amount: int) {  
  when(acc) {    
    acc.balance += amount  
    val id = acc.id()  
    when(log) {  log.record(id, amount) }  
  }  
}
```

R  +

R



-









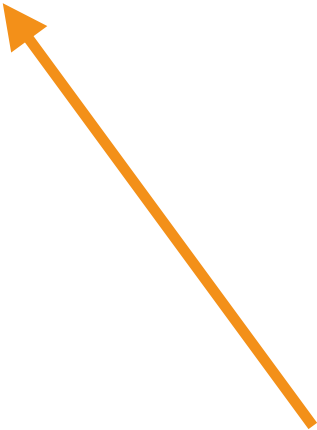
po

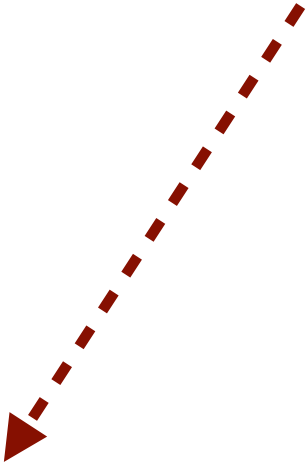
PO



collog







h

b

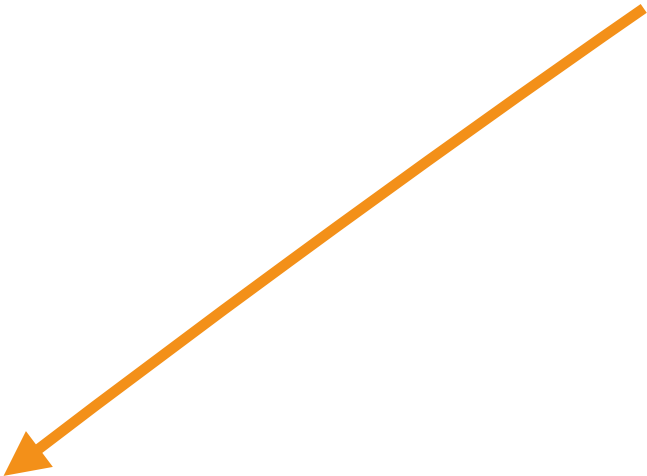








**Intentionally  
disallowed**



COACC









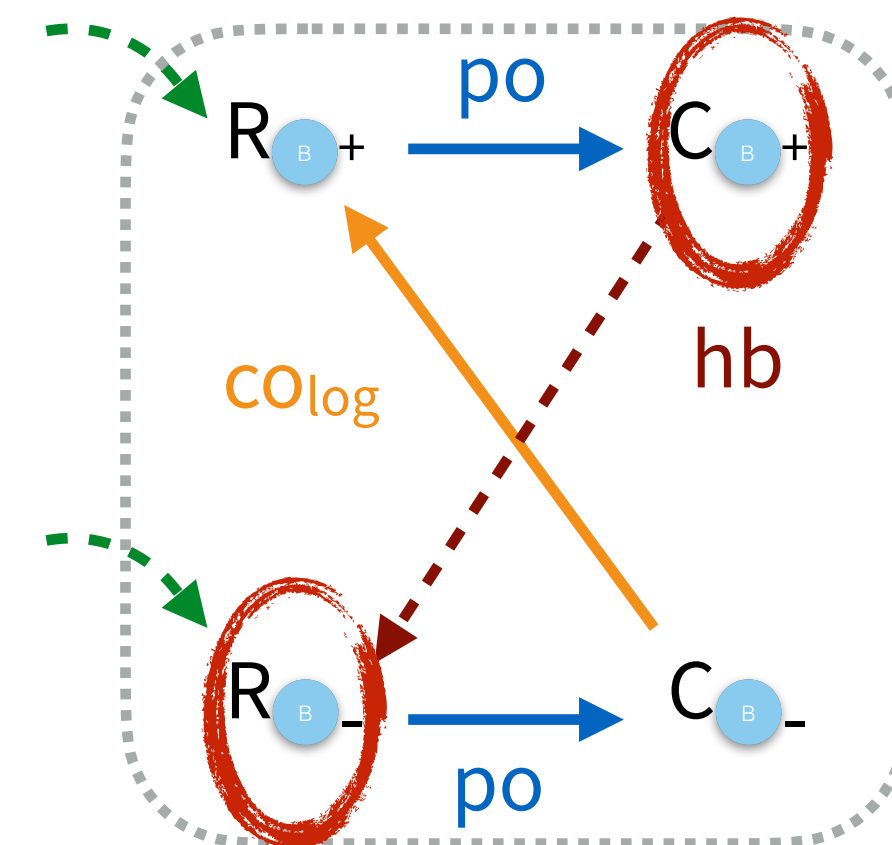
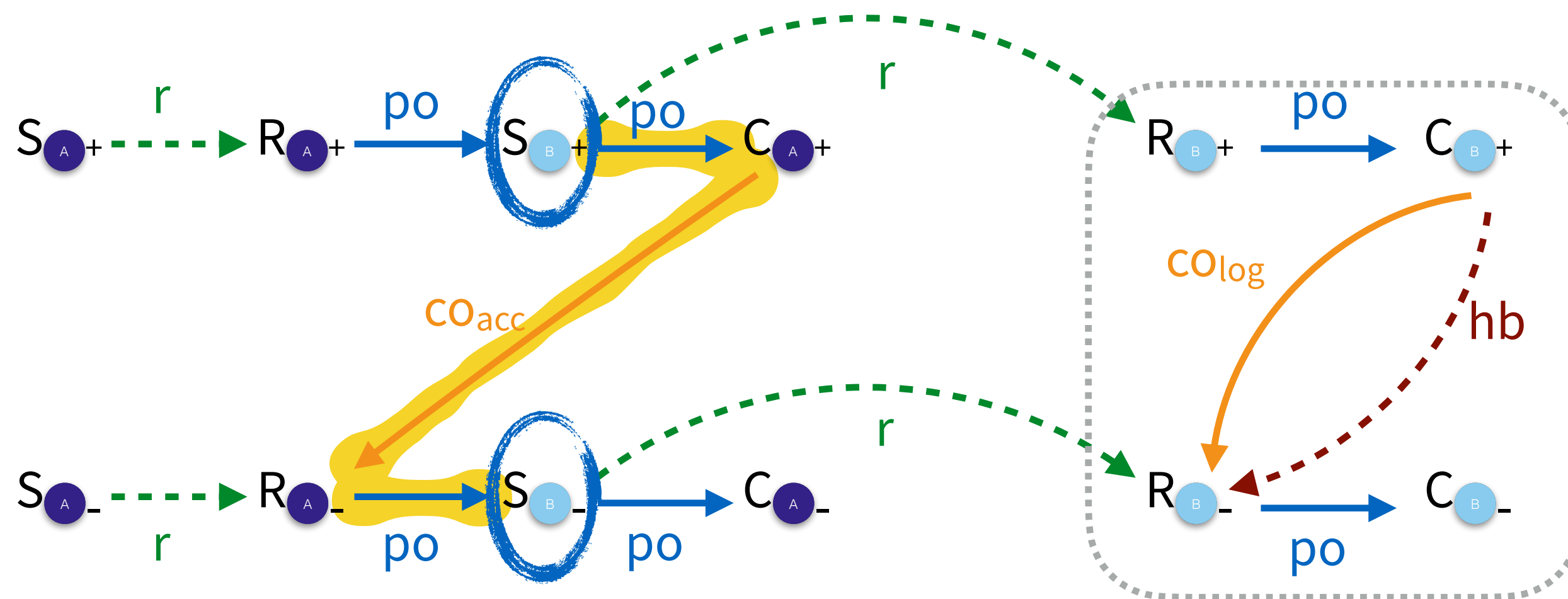


# Validating Causality

```
def update(acc: cown[Account], amount: int) {  
  when(acc) { A  
    acc.balance += amount  
    val id = acc.id()  
    when(log) { B log.record(id, amount) }  
  }  
}
```

update(src, +100)

update(src, -100)



**X** Intentionally disallowed



# What does this get us?

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